

MongoDB Commands:

1.Database & Collection Setup

```
Microsoft Windows [Version 10.0.22631.5624]
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C:\Users\jino>mongosh
Current Mongosh Log ID: 6ac1f185a2fdad195f698e93
Connecting to:      mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.12.0
Using MongoDB:      9.0.2
Using Mongosh:       2.12.0

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
  The server generated these startup warnings when booting
  2026-10-04T19:16:17.261+13:00: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

test> use studentDB
switched to db studentDB
studentDB> db.createCollection("students")
{ ok: 1 }
studentDB> show dbs
admin      40.00 KiB
config     72.00 KiB
local      64.00 KiB
studentDB  8.00 KiB
studentDB> show collections
students
studentDB>
```

2.Insert Operations:

```
studentDB> db.students.insertOne({name:"Aarav", age:21, course:"MERN Stack", status:"active"})
{
  acknowledged: true,
  insertedId: ObjectId('6ac1f32ea2fdad195f698e94')
}
studentDB> db.students.insertMany([{name:"Mia", age:19, course:"MERN" status:"active"},
Uncaught:
SyntaxError: Unexpected token, expected ",", (1:59)

> 1 | db.students.insertMany([{name:"Mia", age:19, course:"MERN" status:"active"},
    |                               ^
    2 |

studentDB> db.students.insertMany([{name:"Mia", age:19, course:"MERN", status:"active"}, {name:"Lio", age:24, course:"DS", status:"a
tive"}, {name:"Sophia",age:22, course:"UI/UX", status:"completed"},{name:"Noah", age:27, course:"MERN", status:"dropped", email:"no
ah@gmail.com"},{name:"Emma", age:20,course:"DS", status:"active"}])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6ac1f54ea2fdad195f698e95'),
    '1': ObjectId('6ac1f54ea2fdad195f698e96'),
    '2': ObjectId('6ac1f54ea2fdad195f698e97'),
    '3': ObjectId('6ac1f54ea2fdad195f698e98'),
    '4': ObjectId('6ac1f54ea2fdad195f698e99')
  }
}
```

3.Read Operations:

```

studentDB> db.students.find()
[
  {
    _id: ObjectId('6ac1f32ea2fdad195f698e94'),
    name: 'AArav',
    age: 21,
    course: 'MERN Stack',
    status: 'active'
  },
  {
    _id: ObjectId('6ac1f54ea2fdad195f698e95'),
    name: 'Mia',
    age: 19,
    course: 'MERN',
    status: 'active'
  },
  {
    _id: ObjectId('6ac1f54ea2fdad195f698e96'),
    name: 'Lio',
    age: 24,
    course: 'DS',
    status: 'active'
  },
  {
    _id: ObjectId('6ac1f54ea2fdad195f698e97'),
    name: 'Sophia',
    age: 22,
    course: 'UI/UX',
    status: 'completed'
  },
  {
    _id: ObjectId('6ac1f54ea2fdad195f698e98'),
    name: 'Noah',
    age: 27,
    course: 'MERN',
    status: 'dropped',
    email: 'noah@gmail.com'
  },
  {
    _id: ObjectId('6ac1f54ea2fdad195f698e99'),
    name: 'Emma',
    age: 20,
    course: 'DS',
    status: 'active'
  }
]

```

```

]
studentDB> db.students.find({course:"MERN"})
[
  {
    _id: ObjectId('6ac1f54ea2fdad195f698e95'),
    name: 'Mia',
    age: 19,
    course: 'MERN',
    status: 'active'
  },
  {
    _id: ObjectId('6ac1f54ea2fdad195f698e98'),
    name: 'Noah',
    age: 27,
    course: 'MERN',
    status: 'dropped',
    email: 'noah@gmail.com'
  }
]
studentDB>

```

4.Update Operations:

```
studentDB> db.students.updateOne({name:"Aarav"}, {$set:{status:"completed"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
```

```
studentDB> db.students.updateMany({course:"DS"},{$set:{status:"completed"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
```

5.Delete Operations:

```
studentDB> db.students.deleteOne({name:"Noah"})
{ acknowledged: true, deletedCount: 1 }
studentDB> db.students.deleteMany({})
{ acknowledged: true, deletedCount: 5 }
studentDB>
```

6.Query Operators:

```
studentDB> db.students.find({age:{$gt:21}})
[
  {
    _id: ObjectId('6ac1fbada2fdad195f698e9c'),
    name: 'Lio',
    age: 24,
    course: 'DS',
    status: 'active'
  },
  {
    _id: ObjectId('6ac1fbada2fdad195f698e9d'),
    name: 'Sophia',
    age: 22,
    course: 'UI/UX',
    status: 'completed'
  },
  {
    _id: ObjectId('6ac1fbada2fdad195f698e9e'),
    name: 'Noah',
    age: 27,
    course: 'MERN',
    status: 'dropped',
    email: 'noah@gmail.com'
  }
]
```

```
studentDB> db.students.find({age:{$lt:21}})
[
  {
    _id: ObjectId('6ac1fbada2fdad195f698e9b'),
    name: 'Mia',
    age: 19,
    status: 'active'
  },
  {
    _id: ObjectId('6ac1fbada2fdad195f698e9f'),
    name: 'Emma',
    age: 20,
    course: 'DS',
    status: 'active'
  }
]
```

```
studentDB> db.students.find({course:{$in:["MERN", "UI/UX"]}})
[
  {
    _id: ObjectId('6ac1fbada2fdad195f698e9b'),
    name: 'Mia',
    age: 19,
    course: 'MERN',
    status: 'active'
  },
  {
    _id: ObjectId('6ac1fbada2fdad195f698e9d'),
    name: 'Sophia',
    age: 22,
    course: 'UI/UX',
    status: 'completed'
  },
  {
    _id: ObjectId('6ac1fbada2fdad195f698e9e'),
    name: 'Noah',
    age: 27,
    course: 'MERN',
    status: 'dropped',
    email: 'noah@gmail.com'
  }
]
```

```
studentDB> db.students.find({$and:[{course:"MERN"}, {status:"active"}]})
[
  {
    _id: ObjectId('6ac1fbada2fdad195f698e9b'),
    name: 'Mia',
    age: 19,
    course: 'MERN',
    status: 'active'
  }
]
studentDB> db.students.find({$or:[{status:"completed"}, {age:{$gt:25}}]})
[
  {
    _id: ObjectId('6ac1fbada2fdad195f698e9d'),
    name: 'Sophia',
    age: 22,
    course: 'UI/UX',
    status: 'completed'
  },
  {
    _id: ObjectId('6ac1fbada2fdad195f698e9e'),
    name: 'Noah',
    age: 27,
    course: 'MERN',
    status: 'dropped',
    email: 'noah@gmail.com'
  }
]
```

```
studentDB> db.students.find({email:{$exists:true}})
[
  {
    _id: ObjectId('6ac1fbada2fdad195f698e9e'),
    name: 'Noah',
    age: 27,
    course: 'MERN',
    status: 'dropped',
    email: 'noah@gmail.com'
  }
]
studentDB> db.students.find({email:{$exists:false}})
[
  {
    _id: ObjectId('6ac1fb9da2fdad195f698e9a'),
    name: 'AArav',
    age: 21,
    course: 'MERN Stack',
    status: 'active'
  },
  {
    _id: ObjectId('6ac1fbada2fdad195f698e9b'),
    name: 'Mia',
    age: 19,
    course: 'MERN',
    status: 'active'
  },
  {
    _id: ObjectId('6ac1fbada2fdad195f698e9c'),
    name: 'Lio',
    age: 24,
    course: 'DS',
    status: 'active'
  },
  {
    _id: ObjectId('6ac1fbada2fdad195f698e9d'),
    name: 'Sophia',
    age: 22,
    course: 'UI/UX',
    status: 'completed'
  },
]
```